Routing & Wireless Concepts Project 2026

Description:

Aurora Healthcare Group, a growing Irish private healthcare provider operating clinics and day-surgery centres, has tasked you with designing and implementing a secure and efficient LAN-based internetwork. The organisation operates from a main Central Campus in Dublin and has two regional clinics located in Limerick and Sligo. Each location requires seamless communication, strong local security controls (especially at Layer 2), and scalability to support future expansion of services and staff.

Your role is to create a prototype network that fulfils the company's requirements and addresses potential vulnerabilities on the LAN segments. This prototype will be presented to Aurora Healthcare Group along with a detailed report explaining the design decisions, network features and how the requirements are met.

This project will give you experience in designing Local Area Networks (LANs), configuring VLANs, addressing LAN security vulnerabilities, implementing IP address allocation and setting up inter-site communication. It will also challenge you to think critically about network scalability, security and reliability in a healthcare environment.

Requirements:

Requirement 1 – VLAN Design at Central Campus

The Dublin Central Campus must include two separate VLANs to segment network traffic and improve performance and security:

- Clinical VLAN (VLAN 100) for 110 users (doctors, nurses and clinical staff).
- Admin VLAN (VLAN 200) for 25 users (HR, finance, management).

End users in the two VLANs should be able to communicate with each other as needed (for example, clinical staff accessing admin systems), using inter-VLAN routing configured on the Central Campus router or a multi-layer switch.

VLANs are not required for the Limerick and Sligo clinics, which will operate with simpler flat LANs. The segmentation provided by VLANs is specific to the Central Campus to support its larger and more complex operational structure.

Requirement 2 – IP Addressing and DHCP

Each VLAN at the Central Campus must have its own unique IPv4 address range that clearly reflects the VLAN number, with dynamic addressing via a DHCP server at the Central Campus. Choose appropriate default gateways on the inter-VLAN routing device.

Central Campus – Clinical VLAN (VLAN 100):

- IP Address Range: 10.10.100.0/25
- Network Address: 10.10.100.0
- Broadcast Address: 10.10.100.127
- Usable Host Range: 10.10.100.1 – 10.10.100.126

Central Campus – Admin VLAN (VLAN 200):

- IP Address Range: 10.10.200.0/27
- Network Address: 10.10.200.0
- Broadcast Address: 10.10.200.31
- Usable Host Range: 10.10.200.1 – 10.10.200.30

Each regional clinic requires its own IPv4 address allocation with at least 30 usable IP addresses, dynamically assigned by a local DHCP service running on its edge routing device (the branch router/MLS acts as the DHCP server for that site):

Limerick Clinic IPv4 Network:

- IP Address Range: 172.16.10.0/27
- Network Address: 172.16.10.0
- Broadcast Address: 172.16.10.31
- Usable Host Range: 172.16.10.1 – 172.16.10.30

Sligo Clinic IPv4 Network:

- IP Address Range: 172.16.20.0/27
- Network Address: 172.16.20.0
- Broadcast Address: 172.16.20.31
- Usable Host Range: 172.16.20.1 – 172.16.20.30

Requirement 3 – LAN Layer 2 Security at Central Campus

The Central Campus LAN must implement Layer 2 security measures on the switches to mitigate the following vulnerabilities:

- MAC Table Attacks: Prevent attackers from flooding the switch MAC address table (e.g. using port security).
- VLAN Attacks: Protect against VLAN hopping and double-tagging to ensure VLAN segmentation is not bypassed.
- STP Attacks: Safeguard the spanning tree topology from malicious BPDUs that could cause loops or force a rogue root bridge.
- DHCP Attacks: Mitigate DHCP starvation and rogue DHCP servers to prevent denial of service or unauthorised configurations.
- ARP Spoofing: Address ARP spoofing to prevent attackers from impersonating devices and intercepting traffic.

Requirement 4 – Secure Remote Administration

Ensure the Central Campus edge routing device (Router or MLS) is remotely accessible by two network administrators using SSH with local authentication. Create two local user accounts with different privilege levels and ensure that remote access via Telnet is disabled.

Requirement 5 – Static and Floating Static Routing Between Sites

Configure static routes to enable communication between:

- Central Campus and Limerick Clinic
- Central Campus and Sligo Clinic
- Limerick and Sligo Clinics (via the Central Campus or via an intermediate WAN router, as per your topology)

The Central Campus and clinics will not be connected to the same router, therefore static routes will be required on the relevant routers to connect the remote networks.

In addition, you must configure floating static routes as backup routes on the appropriate routers. These floating static routes should provide an alternative path between the Central Campus and each clinic (and/or between clinics) in the event of a primary WAN link failure.

You should clearly identify in your design and documentation which links are primary and which are backup, and how the floating static routes support resilience and failover.

Requirement 6 – Wireless Network at Central Campus

The Central Campus must include a basic wireless LAN configuration to support mobile clinical and admin devices:

- Configure a wireless access point (WAP) with:
    - SSID: AuroraHQ-WiFi
    - Use the strongest wireless security mode available in Packet Tracer (for example, WPA2-PSK) for secure access.
- Assign an IP address range 192.168.60.0/24 specifically for wireless clients.
- Ensure wireless clients can access both VLANs (Clinical and Admin) through inter-VLAN routing.
- Implement basic wireless security measures such as MAC filtering for authorised devices and, where supported by the selected wireless device, disabling SSID broadcasting.

Task 1 – Prototype Creation

Use Cisco Packet Tracer to create a fully functional prototype of Aurora Healthcare Group's internetwork that fulfils all requirements outlined above.

Your topology must include at minimum:

- A Central Campus with a Layer 3 device that provides:
    - Inter-VLAN routing for VLAN 100 (Clinical) and VLAN 200 (Admin)
    - Static and floating static routes towards the branch clinics
    - (Optionally) DHCP services for the Central Campus VLANs and the wireless subnet (192.168.60.0/24)

This Layer 3 role may be implemented using either:

    - A multi-layer switch, or
    - A router (router-on-a-stick)
- At least three Layer 2 access switches at the Central Campus, with end devices from both VLAN 100 (Clinical) and VLAN 200 (Admin) distributed across all access switches.
- Redundant Layer 2 links between the Central Campus switches to create a looped topology that requires STP, allowing you to demonstrate STP operation and STP-related Layer 2 security features.
- A wireless access point (AuroraHQ-WiFi) connected into the Central Campus LAN and placed in the 192.168.60.0/24 subnet.
- DHCP for the Central Campus, implemented in one of the following ways:

        - DHCP service running on the Central Campus Layer 3 device (router or multi-layer switch), or
        - A standalone DHCP server connected to the Central Campus LAN, configured with appropriate scopes for VLAN 100, VLAN 200 and the wireless subnet.
- Two remote clinics (Limerick and Sligo), each with:
    - One edge routing device (also acting as the local DHCP server for that clinic's /27 IPv4 network)
    - One Layer 2 switch
    - End devices in the correct IPv4
- WAN links between the Central Campus Layer 3 device and each clinic router (serial or other appropriate links in Packet Tracer), suitable for implementing static and floating static routes.

You may add additional routers, switches, links or servers if you wish, provided they are clearly explained and justified in your report and support the stated requirements.

Task 2 – Documentation

Produce a concise written report that explains your design and shows how you met the requirements. The report should be approximately 1,500 words (4–6 pages including diagrams).

Your report must include:

- Introduction (short)
    - Briefly describe the Aurora Healthcare Group scenario and the overall aim of your network design (2–3 sentences).
- Network topology overview
    - One clearly labelled network diagram showing the Central Campus, both clinics, main devices (Layer 3 device, switches, routers, servers, WAP) and WAN links.
    - A short explanation (1–2 paragraphs) of the overall topology (how the sites connect and where key functions such as routing, DHCP, wireless and security sit).
- Requirements mapping
    - A section that systematically goes through Requirements 1–6, explaining in a few sentences per requirement how your design and configuration fulfil it.
    - You may use a small table (e.g. "Requirement → Devices/Configurations used") to keep this concise.

- LAN and Layer 2 security
    - A brief explanation (few paragraphs) of the main LAN/Layer 2 security features you implemented (e.g. port security, STP protections, DHCP snooping, ARP protection), including which attacks they mitigate.
- Scalability, security and reliability (short reflection)
    - 1–2 short paragraphs describing how your design could scale to more users/sites, how it supports secure operation, and how features like redundant links and floating static routes improve reliability.
- Appendix (optional)
    - Selected screenshots of key configurations (e.g. VLANs, routing, DHCP, wireless, security settings) if you wish to illustrate specific points.

The report should be clear and to the point, focusing on design choices and how they meet the stated requirements, rather than long theoretical explanations.

Submission Details

- Due Date: Monday, 13th April 2026 @ 11:00 am Irish time
- Submission Format: Submit both the report (.pdf) and the Packet Tracer file (.pkt) via email to keara.barrett@setu.ie.
- Interviews/Defence: Project interviews/defence will take place during the week starting 13th April 2026.
- Weighting: This project accounts for 40% of your grade in the "Networking: Wireless and Routing Concepts" module.